

Quarterly Business Review Summary

Drakkar Lite meets all four of Nordheim Online's goals on DigitalOcean Kubernetes. Run rate is **\$123.15/month** at list prices (~\$4.16/day), capped at **\$147.15** with the third node.

What is running: DOKS 1.36 (NYC), HA control plane, 2 nodes autoscaling to 3 · 2 web and 2–5 API pods behind one DigitalOcean Load Balancer fronted by a Kubernetes Gateway · auto-renewing Let's Encrypt HTTPS with an HTTP→HTTPS redirect · Managed PostgreSQL, private VPC only, encrypted, daily backups · tenant isolation enforced by row-level security.

Results against the four goals

Goal	Status	Measured result
Scalability	Met	API CPU reached 842% of request; the HPA went 2 → 5 pods in 15 s , held, and returned to 2 about 60 s after load ended. Nodes autoscale 2 → 3.
Performance	Met	Two 5-minute HTTPS runs at 50 virtual users: p95 132–144 ms against a 300 ms target, p99 229 ms, 39.5–39.9 req/s , 0.00% errors .
Reliability	Met	0 failed requests during a rollout (275), a pod deletion (205) and a node drain (269). Rollback succeeded; PDBs hold ≥ 1 API and ≥ 1 web pod.
Cost optimization	Met	Idle nodes at 5–6% CPU and $\sim\frac{1}{3}$ memory; each API pod ~ 1 millicore / 11 MiB. One load balancer serves both tiers; HTTPS added \$0 .

Recommendations

Nothing that adds cost is enabled without Nordheim Online's approval.

#	Recommendation	\$/mo
R3	Make pod spreading required, not preferred. Both API pods were seen on one node; losing it briefly takes the API down.	0

R5	Add a PostgreSQL standby (2 GiB primary + standby). The database is the only component with no failover.	+45
R6	Enable PgBouncer pooling. The plan allows 22 connections; 5 pods use 15. Required before raising the pod ceiling.	0
R9	Choose a CI/CD approach — scripted manual releases, GitHub Actions, or a self-hosted runner. Not implemented, at your request.	0
R11	Re-review node size after 30 days; s-2vcpu-2gb saves \$12/mo if measured memory allows.	−12

Cost impact: the standby (R5) takes it to **≈\$168/mo**.

Risks and mitigations

Risk	Watch for	Mitigation
Single database node	DB health alerts	Standby (R5); daily backups + PITR
Connection exhaustion	Connections nearing 22	Pods capped at 5 (15 conns); PgBouncer
Spike beyond 5 pods / 3 nodes	HPA at max, p95 rising	Raise limits after R6, within agreed